SAVEETHA SCHOOL OF ENGINEERING

SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

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Reg.no: 192511426 **Course Name:Artificial IntelligenceExp :05 Write the python program for Tic Tac Toe game**

Aim:

To write a Python program to implement a Feed Forward Neural Network.

Algorithm:

Step1: Define the input values representing the features of the dataset.

Step2: Initialize the weights and biases for the connections between the input layer, hidden layer, and output layer.

Step3: Define an activation function, such as the sigmoid function, to introduce non-linearity into the network.

Step4: Calculate the weighted sum of inputs and biases for the hidden layer nodes.

Step5: Apply the activation function to the hidden layer sum to generate the hidden layer outputs.

Step6: Calculate the weighted sum of the hidden layer outputs and output biases, then apply the activation function to generate the final network prediction.

Program:

```
import math


def sigmoid(x):

    return 1 / (1 + math.exp(-x))


def feed_forward(inputs, weights_in_hid, bias_hid, weights_hid_out, bias_out):

    hidden_output = []


    for i in range(len(weights_in_hid[0])):
```

```
        activation = bias_hid[i]
        for j in range(len(inputs)):
            activation += inputs[j] * weights_in_hid[j][i]
        hidden_output.append(sigmoid(activation))

    final_output = []

    for i in range(len(weights_hid_out[0])):
        activation = bias_out[i]
        for j in range(len(hidden_output)):
            activation += hidden_output[j] * weights_hid_out[j][i]
        final_output.append(sigmoid(activation))

    return hidden_output, final_output

if __name__ == '__main__':
    input_values = [0.5, 0.8]

    w_input_hidden = [
        [0.2, 0.4],
        [0.3, 0.5]
    ]
    b_hidden = [0.1, 0.1]

    w_hidden_output = [
        [0.6],
        [0.7]
    ]
    b_output = [0.2]
```

```
hidden_res, final_res = feed_forward(input_values, w_input_hidden, b_hidden, w_hidden_output, b_output)

print("Input Values:")
print(input_values)
print("Hidden Layer Outputs:")
for val in hidden_res:
    print(round(val, 4))
print("Final Network Output:")
for val in final_res:
    print(round(val, 4))
```

OUTPUT:

```
Input Values:
[0.5, 0.8]
Hidden Layer Outputs:
0.6083
0.6682
Final Network Output:
0.7374
```

Result: Thus, the program was executed and found that the output was found to be correct.